

User instructions

Model: CS-MCQ19242100
CS-MCQ28241100



Shenzhen Coolingstyle Technology Co., Ltd.

Content

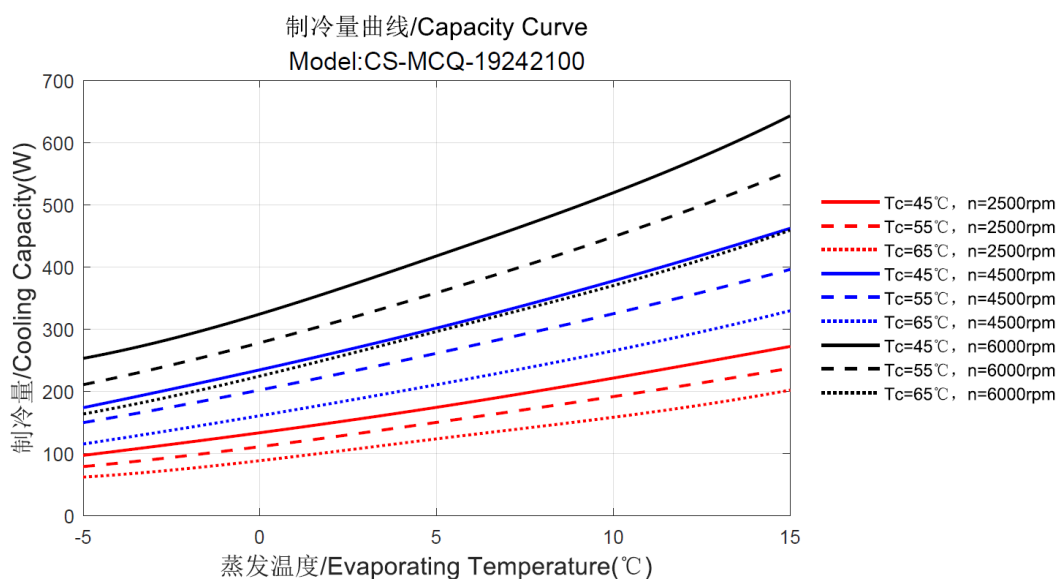
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1. Basic parameters

Name	Specification	
	CS-MCQ-19242100	CS-MCQ-28241100
Compressor Type	Rotary DC Timing Compressors	
Rated Voltage	DC 24V	
Voltage range	24V $\pm$ 10%	
Rated Current	10A	12A
Maximum Current	12A	16A
Displacement	1.9cm ³ /rev	2.8 cm ³ /rev
Motor Rotor Pole	6	
Maximum Compression Ratio	8	6
Speed Range	1200~6300rpm	1200~5500rpm
Max Cooling Capacity	440W	550W
Refrigerant	R134a	
Oil	RL68H	
Oil Charge	35mL	50mL
Ambient Temperature	-5~55℃	
Evaporating Temperature	-10~15℃	
Condensing Temperature	45~76℃	
Weight (Compressor Body Only)	0.85kg	1.3kg
Suction Tube(I.D)	6.5mm	
Discharge Tube(I.D)	6.5mm	

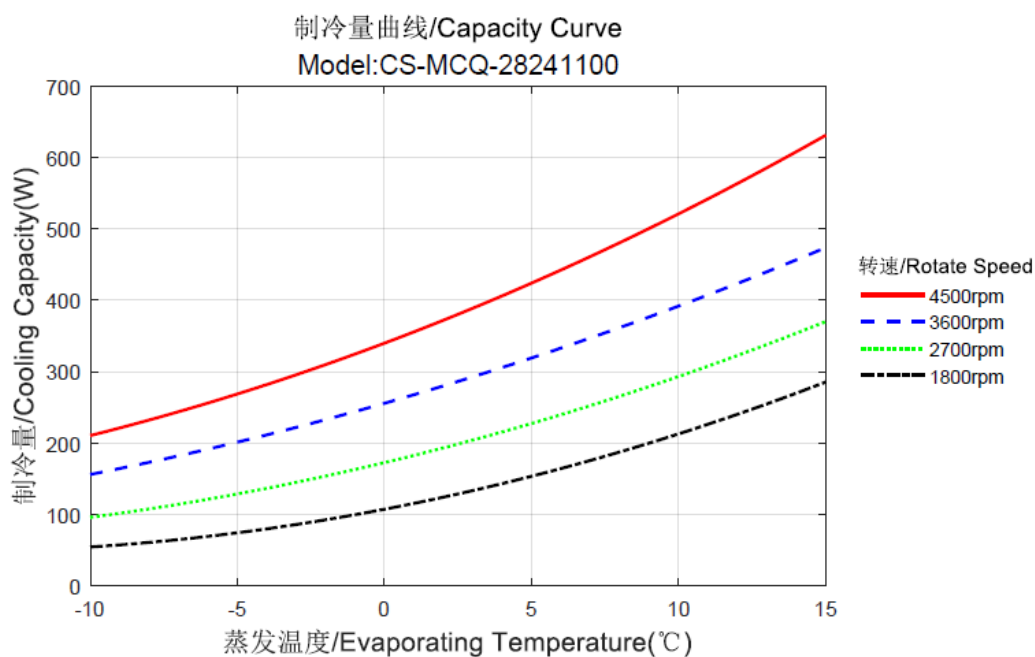
2. Cooling capacity curve

CS-MCQ-19242100 Cooling capacity curve



Tc: Condensation temperature ; n: Compressor speed

CS-MCQ-28241100 Cooling capacity curve

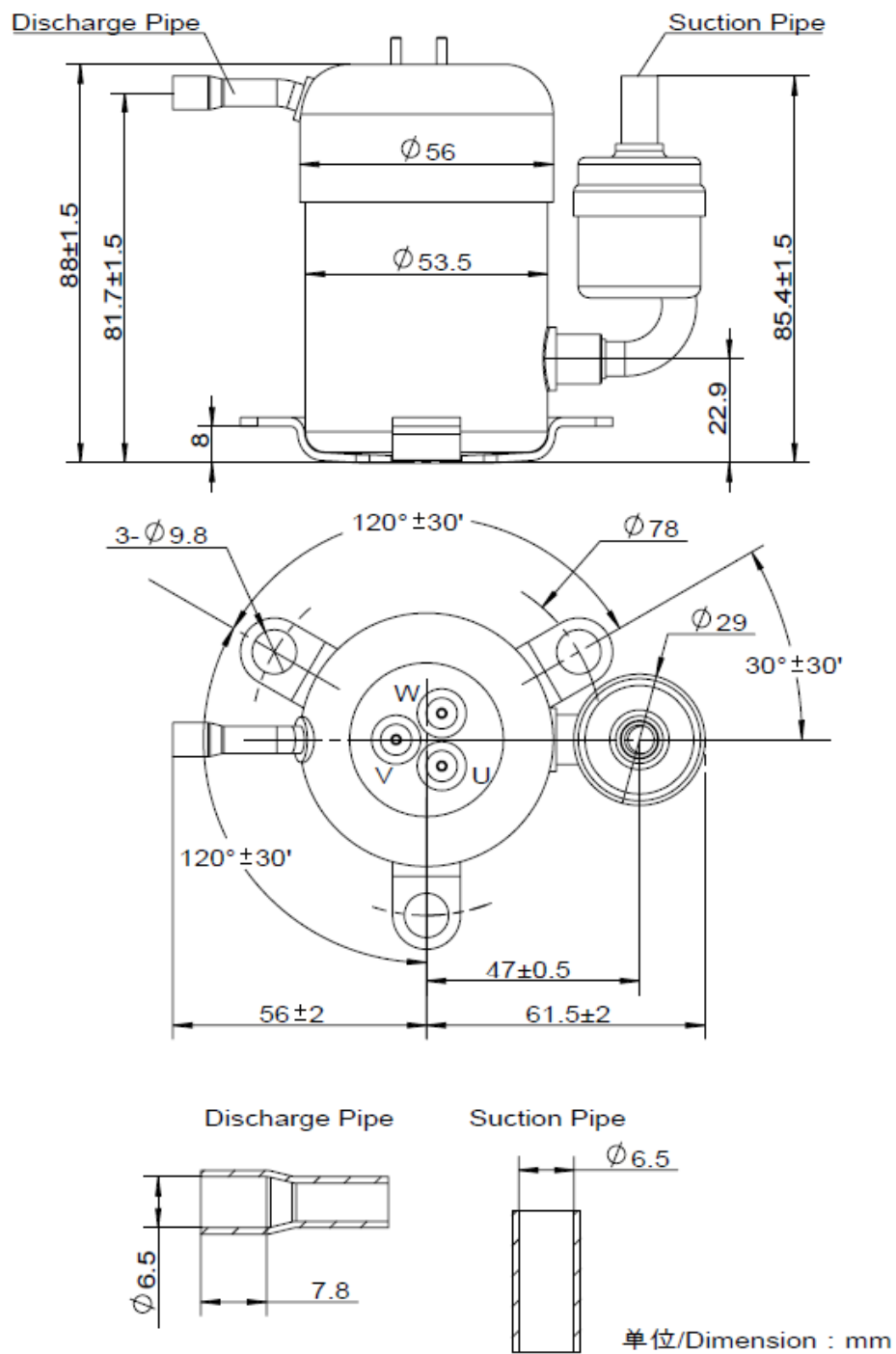


Test conditions: condensation temperature: 54.4°C; subcooling: 8.3°C; ambient temperature: 35°C

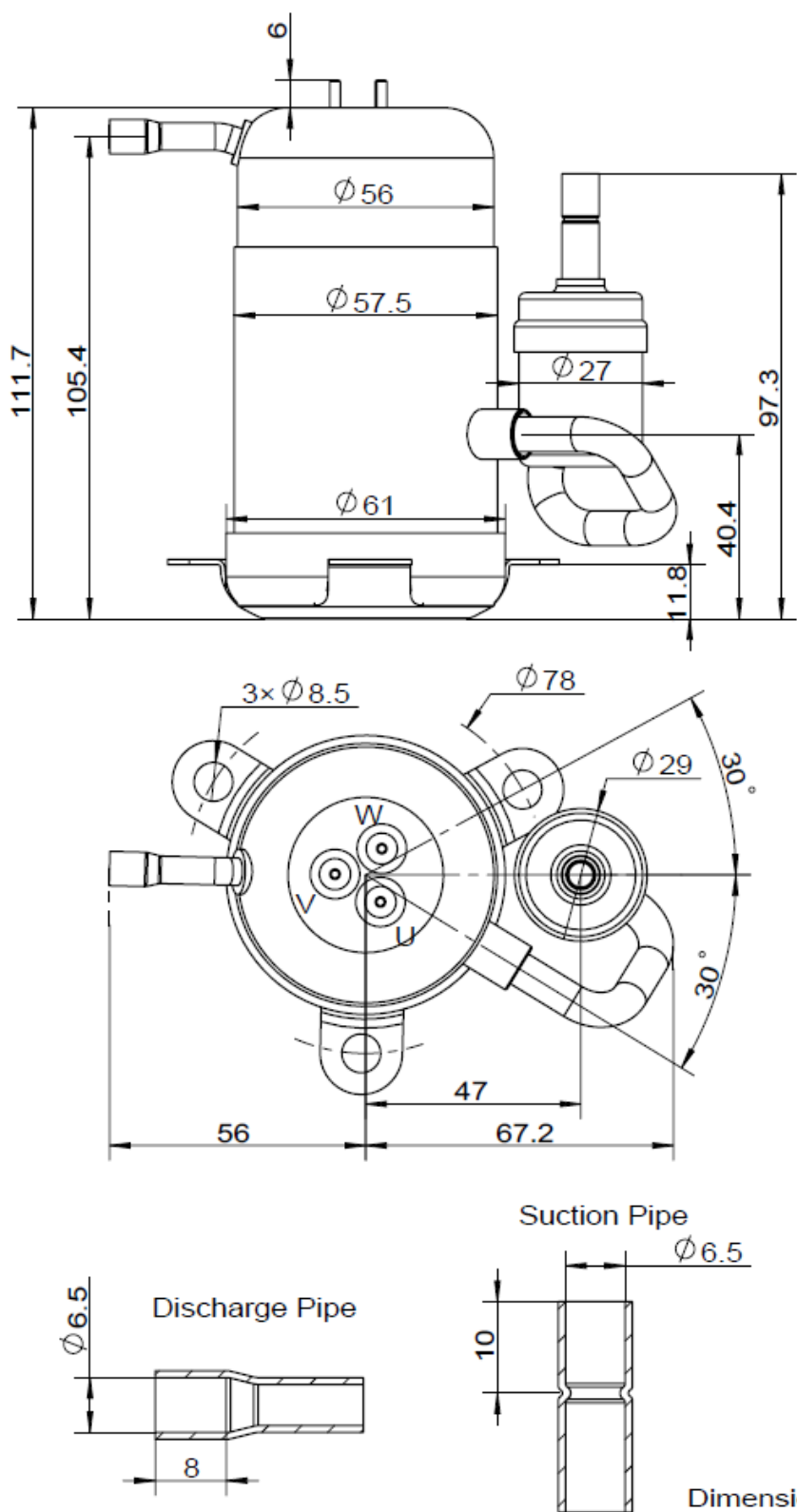
3. Size

3.1 Compressor size

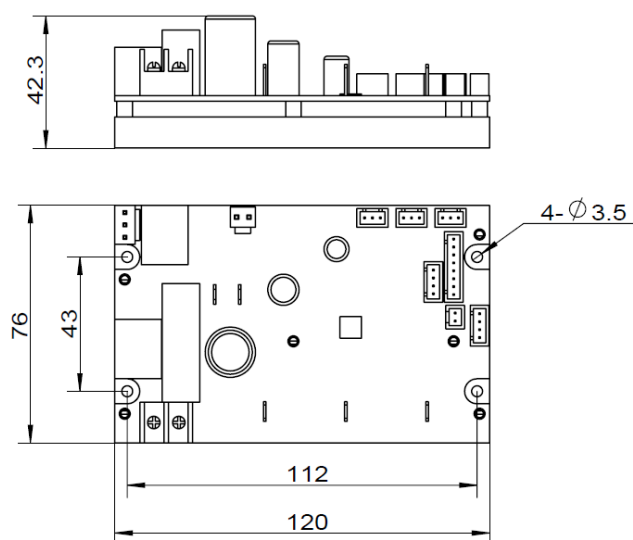
CS-MCQ-19242100 compressor size



CS-MCQ-28241100 compressor size



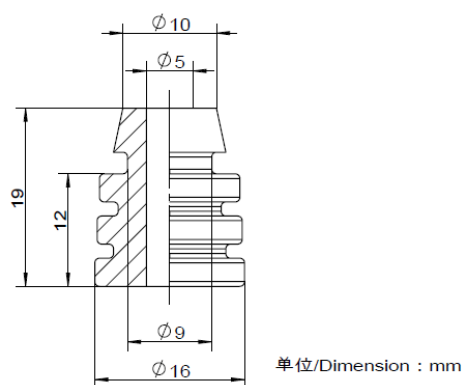
3.2 Drive board



单位/dimension : mm

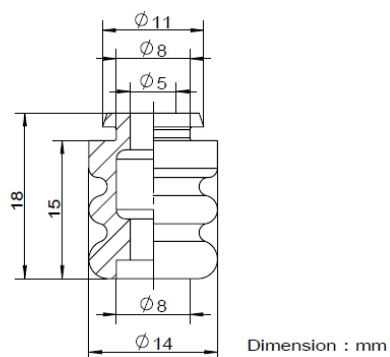
3.3 Cushion size

CS-MCQ-19242100 cushion size



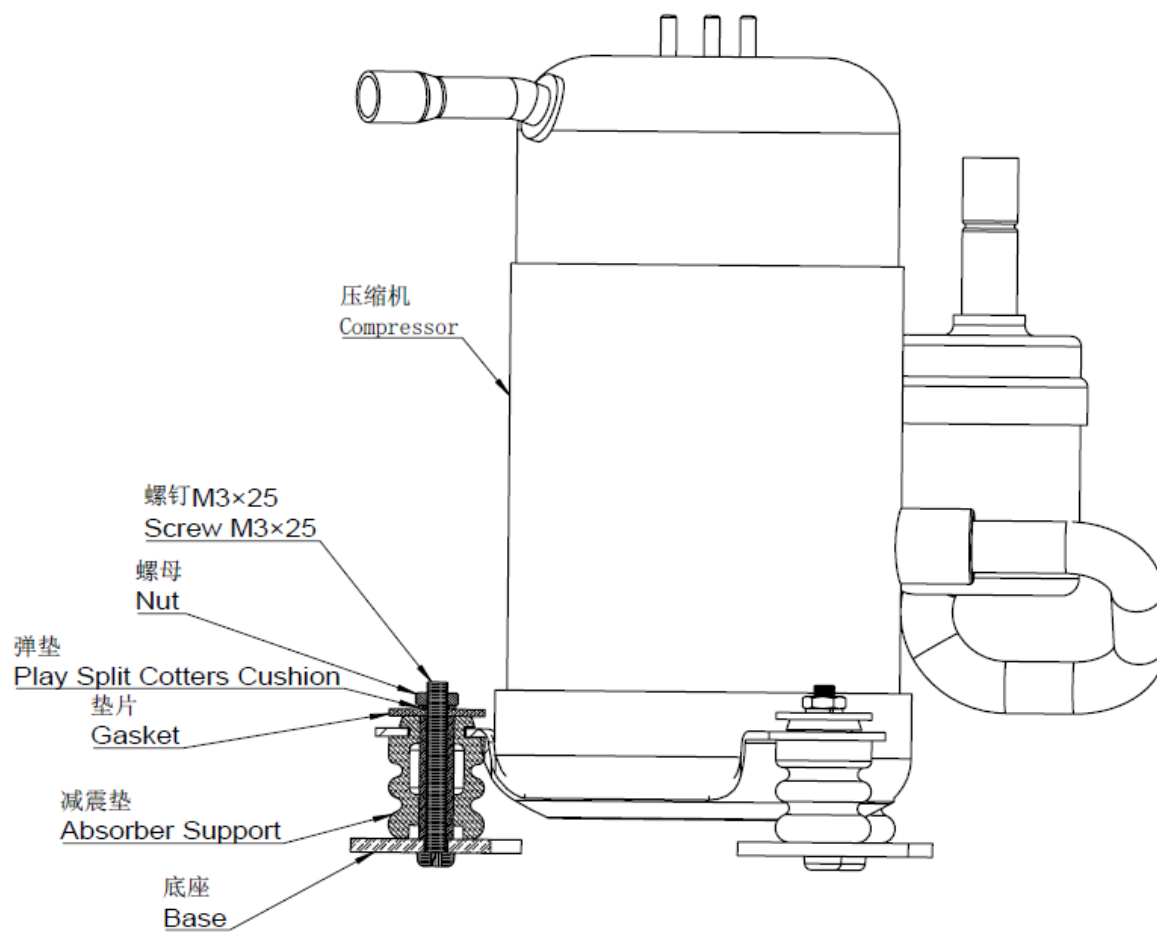
单位/Dimension : mm

CS-MCQ-28241100 cushion size



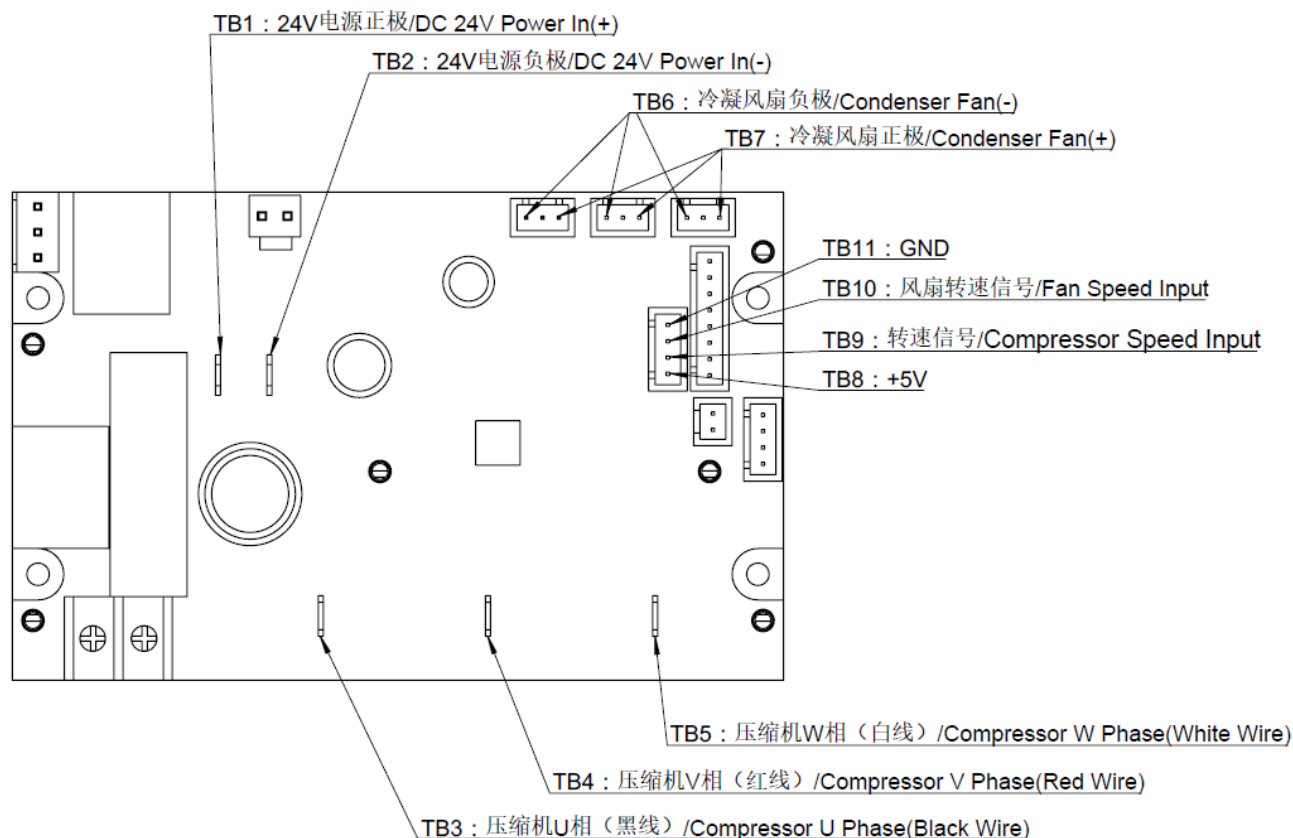
Dimension : mm

3.4 Installation



4. Driver wiring instructions

MCQ series compressors must be driven by matching drivers. The driver interface is shown in the figure below.



4.1 Power input port (TB1, TB2)

The driver input voltage range is $24V \pm 10\%$, port TB1 is positive and TB2 is negative.

4.2 Wiring ports of driver and compressor (TB3, TB4, TB5)

The driver and compressor are connected by a dedicated three-core connecting cable. The colors of the three-core cable are black, red and white. The black wire is connected to port TB3; the red wire is connected to port TB4; the white wire is connected to port TB5. The other end of the three-core wire has been connected to the terminal block, and the terminal block is fastened to the three terminals on the top of the compressor, so there is no need to distinguish the orientation.

4.3 Condensing fan power supply port (TB6, TB7)

The driver is equipped with ports TB6 and TB7 specifically for powering the condensing fan, where TB6 is the negative pole and TB7 is the positive pole. There are 3 fan ports. This port only has voltage output when the compressor is running. The output voltage is 24V, and the maximum load of each port is 1A. Between TB6 and TB7 is the fan speed control port, which is only useful for speed control fans.

4.4 5V voltage output port (TB8)

When the driver is connected to the power supply, port TB8 can output a 5V voltage, which can be used to adjust the compressor speed. Details will be introduced later.

4.5 Compressor speed signal input port (TB9)

TB9 is the signal input port for controlling the compressor speed and can receive voltage signals in the range of 0~5V (relative to GND). The corresponding relationship between voltage value and compressor speed is shown in the table below.

Speed voltage U/V	Compressor speed N/rpm	
	CS-MCQ-19242100	CS-MCQ-28241100
0~1	0	0
1	1200	1200
$1 < U < 4.65$	$1397 * (U - 1) + 1200$	$1178 * (U - 1) + 1200$
$4.65 \leq U \leq 5$	6300	5500

4.6 Fan speed signal input port (TB10)

TB10 is used to adjust the fan speed. When using it, you need to refer to the specifications of the fan used. It will not be used under normal circumstances.

4.7 GND (TB11)

TB11 is mainly used to connect the control signal of the compressor. All voltage signals on the driver are relative to GND. If an external circuit is used to control the compressor, the external circuit must also be grounded to the driver.

4.8 Driver indicator light description

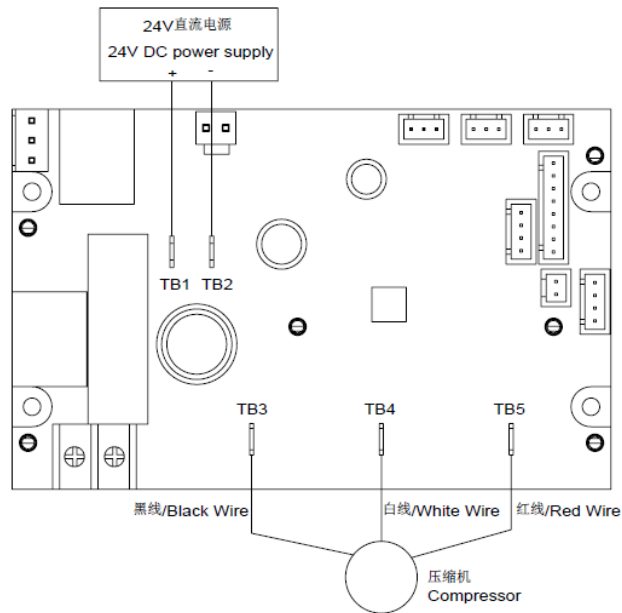
There is an indicator light on the drive. The lighting status of the indicator light can reflect the working or fault status of the compressor. The specific description is as shown in the table below.

Working/fault Status	Illustrate	Status of indicator light
Standby	The drive is connected to the power supply, but the compressor does not start (the speed voltage is lower than the starting voltage).	Flash slowly
Normal work		Flash slowly
Driver overcurrent protection	The input current of TB1 and TB2 exceeds the protection value.	Fast and steady flashing
Overvoltage protection	When the driver input voltage is higher than 32V, the high voltage protection is triggered. When the voltage drops below 29V and remains for 5 seconds, the driver returns to normal and the compressor restarts.	Flashes 2 times in a loop
Under voltage protection	When the driver input voltage is lower than 20V, the low-voltage protection is triggered. When the voltage rises to above 23V and remains for 5 seconds, the driver returns to normal and the compressor restarts.	Flashes 3 times in a loop
Stall protection	The motor inside the compressor is working abnormally.	Flashes 5 times in a loop
Compressor current protection	The output current of TB3, TB4 and TB5 exceeds the protection value.	Flashes 7 times in a loop
Compressor phase loss	One or more of U, V, and W are disconnected from the compressor. The driver will restart the compressor every 30 seconds and will not restart after three consecutive failed restarts.	Flashes 9 times in a loop

5. Typical wiring examples

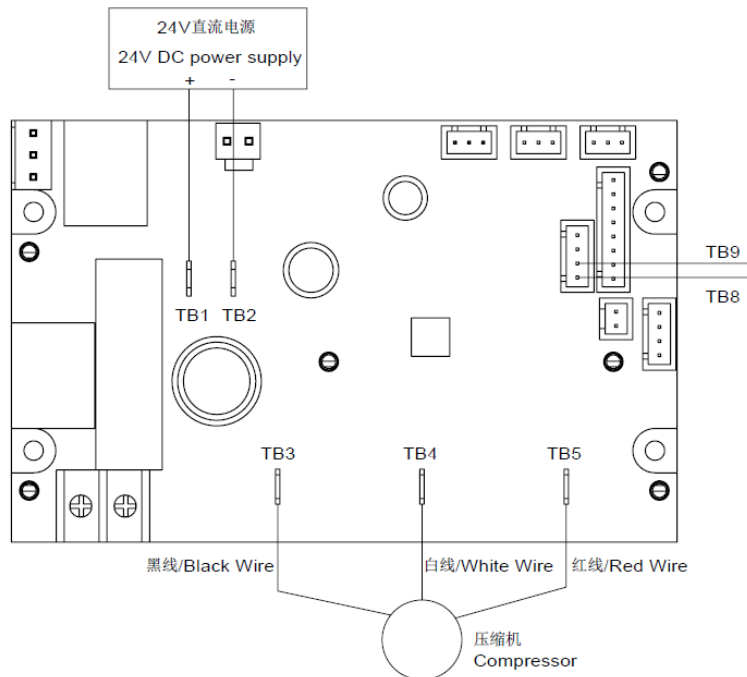
5.1 Example 1

The compressor speed signal TB9 is suspended. At this time, the compressor rotates at a constant speed of approximately 3000 rpm.



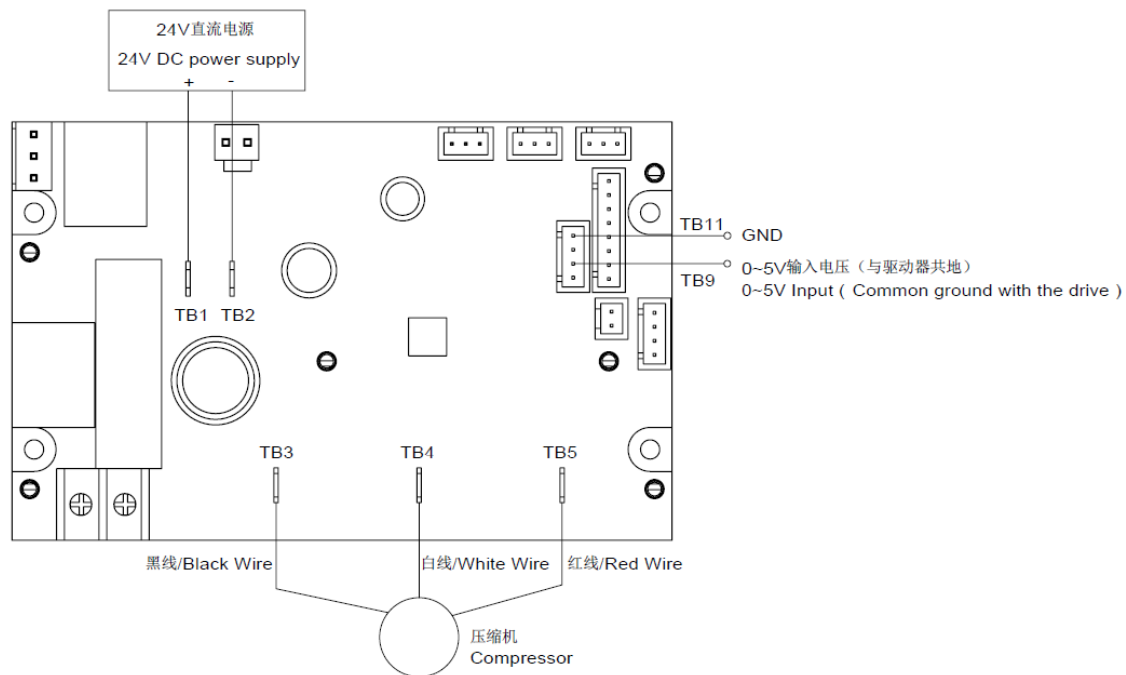
5.2 Example 2

Short-circuit TB8 (+5V) and TB9 (SP), the compressor will run at full speed.



5.3 Example 3

Users can use an external 0~5V voltage signal to control the compressor speed, as shown in the figure below. This voltage signal must be grounded to the driver. See Section 4.5 for the relationship between voltage and speed.



6. Factory standard list

No.	Item	Unit	Qty
1	Compressor	pcs	1
2	Driver	pcs	1
3	Wires	pcs	1
4	Cushion	Pcs	3
5	Metal pillar	Pcs	3
6	Gasket	Pcs	3

7. Notice

1. The compressor needs to be placed horizontally during operation. The maximum inclination

angle cannot exceed 15°, and it cannot be tilted for a long time.

2. The compressor must not be tilted or turned upside down during storage and transportation to prevent lubricating oil from spilling.

3. The compressor has been pre-installed with RL68H lubricating oil when leaving the factory, which is sufficient for general refrigeration systems. For systems that have been charged and discharged with refrigerant multiple times, an appropriate amount of lubricating oil of the same type can be added as appropriate.

4. R134a must be used as refrigerant.

5. When working, the drive should be installed in a location that is easy to dissipate heat.

6. Under any operating conditions, the compression ratio of the compressor should be less than its maximum allowable value.

7. The pressure difference ΔP between the high and low pressure chambers when the compressor is started (discharge chamber pressure - suction chamber pressure) should be below 0.03MPa. After the compressor stops, you need to wait at least 30 seconds for the pressure to balance before starting again.

8. To protect the compressor, neither the rising nor falling speed of the rotation speed exceeds 120rpm/s.

9. The driver should stop working immediately when overcurrent occurs.

10. Before charging, the internal pressure of the refrigeration system should be lower than 20Pa, and the water content should be lower than 40ppm. There should be no impurities, metal powders, acids, alkalis, oils and cleaning agents left in the manufacturing process inside the refrigeration accessories and pipelines. During brazing, protective gas should be injected into the system to prevent the occurrence of oxide scale.

11. It is absolutely prohibited to operate the compressor with air or a mixture of air and working

fluid, as this may cause an explosion.

12. The suction and exhaust pipes of the compressor are blocked with blind plugs and are only allowed to be removed when in use. After removing the blind plug, the inside of the compressor should not be exposed to the atmosphere for more than 5 minutes.

13. Before removing the blind plug, you can run the compressor directly for a short time to observe whether it is normal. After the refrigeration system is evacuated, the compressor cannot be operated directly if it has not been filled with refrigerant.

14. Wiring must be carried out in strict accordance with the wiring instructions. The three phase wires U, V, and W of the compressor must not be connected incorrectly. If connected incorrectly, the compressor may reverse, and the reversed compressor can no longer be used.

15. Do not keep the speed signal below 1.4V for a long time. In this range, the compressor may be damaged due to repeated starts and stops.



Shenzhen Coolingstyle Technology Co., Ltd.

904, Building C, Gangzhilong Technology Park Business Center, No. 6 Qinglong Road, Qinghua
Community, Longhua Street, Longhua District, Shenzhen City, Guangdong ProvinceTel: 0755-
29952987

Website: www.coolingstyle.com

Email: info@coolingstyle.com